



Date: 11/12/2025

Lab Practical #2:

- 1) Write a program to reverse a string using pointers.
- 2) Implement string functions like strlen, strcat, and strcmp.
- 3) Implement a program to read and write a text file.
- 4) Write a C program to test whether a number is prime or composite.
- 5) Write a C program to generate prime numbers in a given range.

Program-1:

```
#include <stdio.h>
#include <string.h>

void reverseString(char *str) {
    char *start = str;
    char *end = str + strlen(str) - 1;
    char temp;

    while (start < end) {
        temp = *start;
        *start = *end;
        *end = temp;
        start++;
        end--;
    }
}

int main() {
    char str[100];
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
    str[strcspn(str, "\n")] = '\0';
    printf("Original string: %s\n", str);
    reverseString(str);
    printf("Reversed string: %s\n", str);
    return 0;
}
```

Output:

```
> gcc 1.c && ./a.out
Enter a string: Abc
Original string: Abc
Reversed string: cbA
```



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Program-2:

```
#include <stdio.h>

int myStrlen(const char *str) {
    int len = 0;
    while (*str++) len++;
    return len;
}

void myStrcat(char *dest, const char *src) {
    while (*dest) dest++;
    while ((*dest++ = *src++));
}

int myStrcmp(const char *str1, const char *str2) {
    while (*str1 && *str2 && *str1 == *str2) { str1++; str2++; }
    return *str1 - *str2;
}

int main() {
    char str1[100], str2[100];
    printf("Enter first string: ");
    fgets(str1, sizeof(str1), stdin);
    str1[myStrlen(str1) - 1] = '\0';
    printf("Enter second string: ");
    fgets(str2, sizeof(str2), stdin);
    str2[myStrlen(str2) - 1] = '\0';
    printf("Length of first string: %d\n", myStrlen(str1));
    printf("Length of second string: %d\n", myStrlen(str2));
    printf("Comparison result: %d\n", myStrcmp(str1, str2));
    myStrcat(str1, " ");
    myStrcat(str1, str2);
    printf("Concatenated string: %s\n", str1);
    return 0;
}
```

Output:

```
> gcc 2.c && ./a.out
Enter first string: Java
Enter second string: JavaScript
Length of first string: 4
Length of second string: 10
Comparison result: -83
Concatenated string: Java JavaScript
```



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Program-3:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    FILE *f;
    char fn[100], t[1000];
    int c;

    scanf("%s%d", fn, &c);

    if (c == 1) {
        f = fopen(fn, "w");
        if (!f) return 1;

        getchar();
        while (fgets(t, sizeof(t), stdin) && strcmp(t, "END\n"))
            fputs(t, f);

        fclose(f);
    }
    else if (c == 2) {
        f = fopen(fn, "r");
        if (!f) return 1;

        while (fgets(t, sizeof(t), f))
            printf("%s", t);

        fclose(f);
    }

    return 0;
}
```

Output:

```
> gcc 3.c && ./a.out
demo.txt
1
INS
END
~/Desktop/sem-6/ins/lab-2 main*
> cat demo.txt
INS
```



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Program-4:

```
#include <stdio.h>
#include <math.h>

int isPrime(int n) {
    if (n <= 1) return 0;
    if (n == 2) return 1;
    if (n % 2 == 0) return 0;
    for (int i = 3; i <= sqrt(n); i += 2)
        if (n % i == 0) return 0;
    return 1;
}

int main() {
    int n;
    scanf("%d", &n);
    printf(isPrime(n) ? "%d is a PRIME number.\n" : "%d is a COMPOSITE number.\n", n);
    return 0;
}
```

Output:

```
) gcc 4.c -lm && ./a.out
67
67 is a PRIME number.
```



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Program-5:

```
#include <stdio.h>
#include <math.h>

int isPrime(int n) {
    if (n <= 1) return 0;
    if (n == 2) return 1;
    if (n % 2 == 0) return 0;
    for (int i = 3; i <= sqrt(n); i += 2)
        if (n % i == 0) return 0;
    return 1;
}

int main() {
    int s, e, c = 0;
    scanf("%d%d", &s, &e);
    for (int i = s; i <= e; i++)
        if (isPrime(i)) {
            printf("%d ", i);
            if (++c % 10 == 0) printf("\n");
        }
    if (c) printf("\nTotal prime numbers: %d\n", c);
    else printf("No prime numbers found in this range.\n");
    return 0;
}
```

Output:

```
> gcc 5.c -lm && ./a.out
16 90
17 19 23 29 31 37 41 43 47 53
59 61 67 71 73 79 83 89
Total prime numbers: 18
```